Heart Dataset

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
2	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
3	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
4	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
5	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
6	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
7	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
8	44	1	1	120	263	0	1	173	0	0	2	0	3	1
9	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
10	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

Dataset Summary: Total Rows and Variables

The CONTENTS Procedure

Data Set Name	WORK.HEART	Observations	303
Member Type	DATA	Variables	14
Engine	V9	Indexes	0
Created	2026-03-01 02:25:56	Observation Length	112
Last Modified	2026-03-01 02:25:56	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1168
Obs in First Data Page	303
Number of Data Set Repairs	0
Filename	/saswork/SAS_work19D700017519_odaws02-usw2.oda.sas.com/SAS_work6B0700017519_odaws02-usw2.oda.sas.com/heart.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	536871069
Access Permission	rw-r--r--
Owner Name	u64430375
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	age	Num	8	BEST12.	BEST32.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
12	ca	Num	8	BEST12.	BEST32.
5	chol	Num	8	BEST12.	BEST32.
3	cp	Num	8	BEST12.	BEST32.
9	exang	Num	8	BEST12.	BEST32.
6	fbs	Num	8	BEST12.	BEST32.
10	oldpeak	Num	8	BEST12.	BEST32.
7	restecg	Num	8	BEST12.	BEST32.
2	sex	Num	8	BEST12.	BEST32.
11	slope	Num	8	BEST12.	BEST32.
14	target	Num	8	BEST12.	BEST32.
13	thal	Num	8	BEST12.	BEST32.
8	thalach	Num	8	BEST12.	BEST32.
4	trestbps	Num	8	BEST12.	BEST32.

Summary Statistics for Heart Dataset

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
age	303	54.3663366	9.0821010	29.0000000	77.0000000
sex	303	0.6831683	0.4660108	0	1.0000000
cp	303	0.9669967	1.0320525	0	3.0000000
trestbps	303	131.6237624	17.5381428	94.0000000	200.0000000
chol	303	246.2640264	51.8307510	126.0000000	564.0000000
fbs	303	0.1485149	0.3561979	0	1.0000000
restecg	303	0.5280528	0.5258596	0	2.0000000
thalach	303	149.6468647	22.9051611	71.0000000	202.0000000
exang	303	0.3267327	0.4697945	0	1.0000000
oldpeak	303	1.0396040	1.1610750	0	6.2000000
slope	303	1.3993399	0.6162261	0	2.0000000
ca	303	0.7293729	1.0226064	0	4.0000000
thal	303	2.3135314	0.6122765	0	3.0000000
target	303	0.5445545	0.4988348	0	1.0000000

Standardized Data (First 10 Rows)

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	1	1
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	2	1
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	2	1
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	2	1
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	2	1
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	1	1
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	2	1
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	3	1
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	3	1
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	2	1

Standardized Data (First 10 Rows)

The FASTCLUS Procedure

Replace=FULL Radius=0 Maxclusters=2 Maxiter=1

Criterion Based on Final Seeds =	0.9480

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	70	0.8834	4.9783		2	1.7293
2	233	0.9648	4.9104		1	1.7293

Pseudo F Statistic =	35.92

Observed Over-All R-Squared =	0.10661

Approximate Expected Over-All R-Squared =	0.15568

Cubic Clustering Criterion =	-6.430

WARNING: The two values above are invalid for correlated variables.

Standardized Data (First 10 Rows)

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=3 Maxiter=1

Criterion Based on Final Seeds =	0.8540

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	98	0.9284	4.0910		2	2.2888
2	194	0.8019	3.4055		1	2.2888
3	11	0.8628	2.9226		1	2.8126

Pseudo F Statistic =	60.53

Observed Over-All R-Squared =	0.28751

Approximate Expected Over-All R-Squared =	0.27577

Cubic Clustering Criterion =	1.330

WARNING: The two values above are invalid for correlated variables.

Standardized Data (First 10 Rows)

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=4 Maxiter=1

Criterion Based on Final Seeds =	0.8142

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	106	0.8510	4.9563		4	1.9099
2	114	0.7189	2.9448		1	2.2757
3	5	0.8718	2.8725		4	3.3551
4	78	0.8322	4.1120		1	1.9099

Pseudo F Statistic =	58.08

Observed Over-All R-Squared =	0.36818

Approximate Expected Over-All R-Squared =	0.37370

Cubic Clustering Criterion =	-0.608

WARNING: The two values above are invalid for correlated variables.

Standardized Data (First 10 Rows)

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=5 Maxiter=1

Criterion Based on Final Seeds =	0.8067

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	69	0.7120	2.8923		2	2.0399
2	123	0.7683	3.7394		3	1.8613
3	89	0.8131	3.1093		2	1.8613
4	21	0.9205	3.4791		3	2.7002
5	1	.	0		2	5.9526

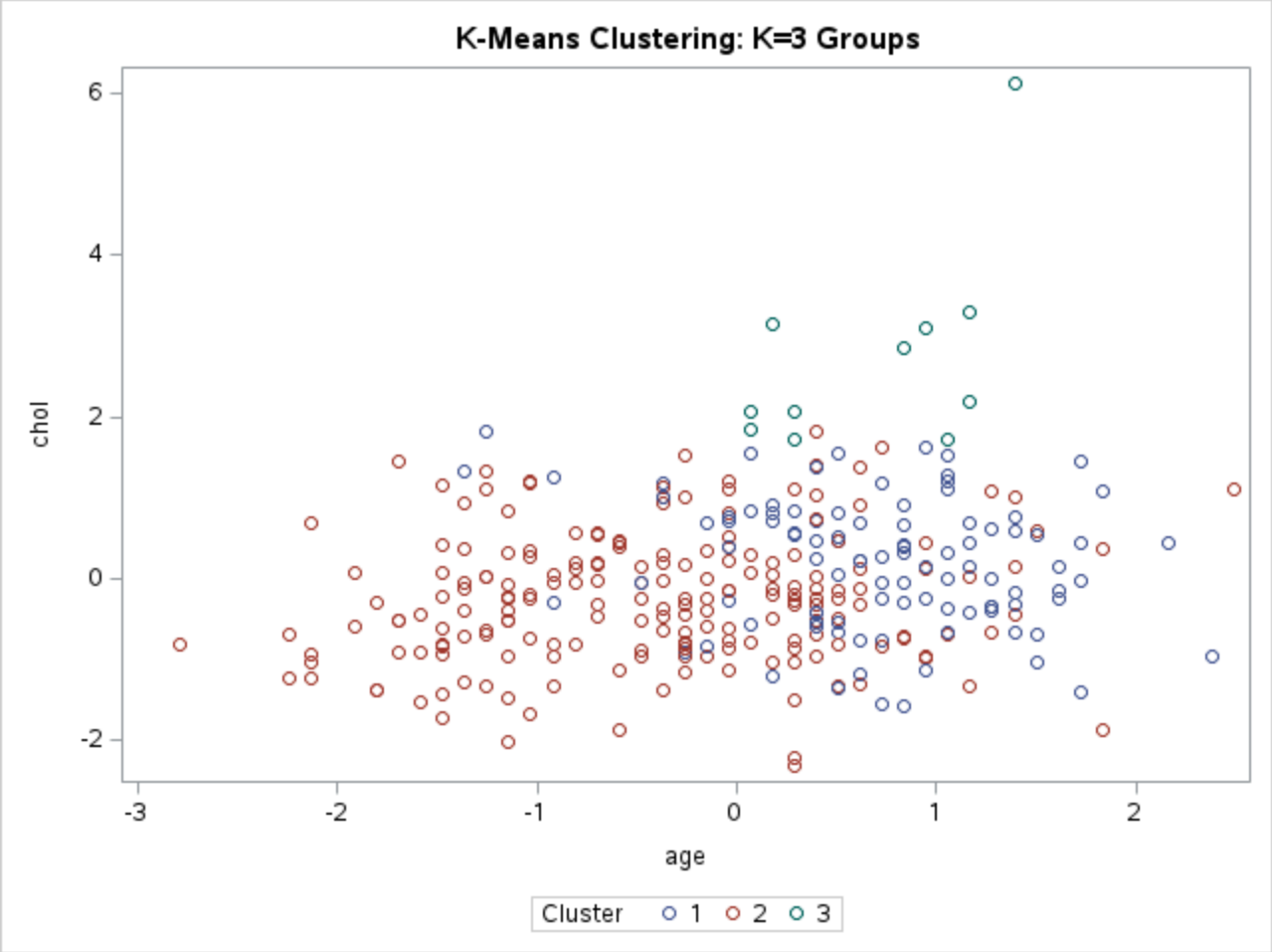
Pseudo F Statistic =	49.38

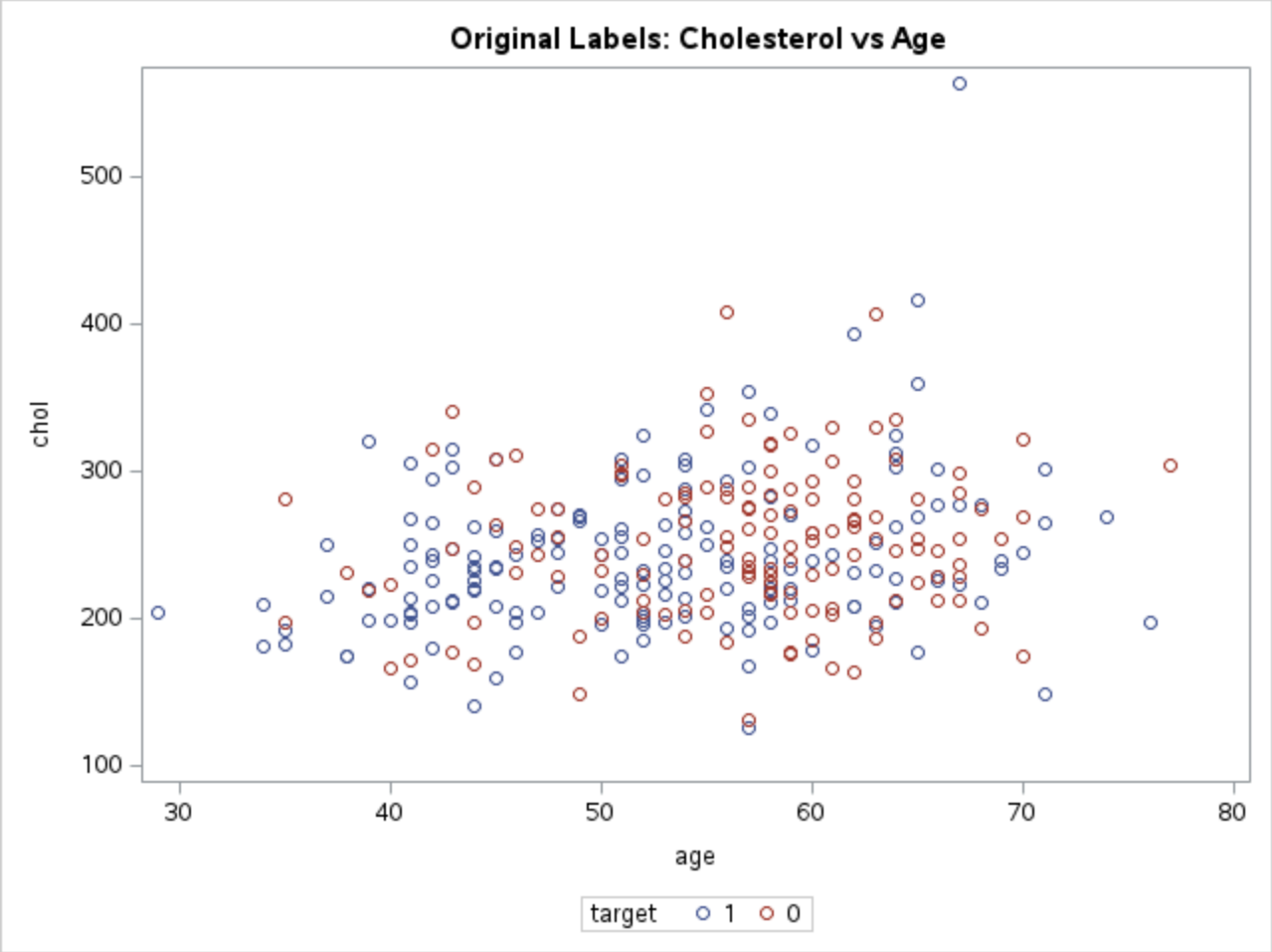
Observed Over-All R-Squared =	0.39863

Approximate Expected Over-All R-Squared =	0.45605

Cubic Clustering Criterion =	-6.321

WARNING: The two values above are invalid for correlated variables.


K-Means Clustering: K=3 Groups
chol
6
4
2
0
-2
-3
-2
-1
0
1
2
age
Cluster
1
2
3


Original Labels: Cholesterol vs Age
chol
500
400
300
200
100
30
40
50
60
70
80
age
target
1
0